Literature Survey: Hallucination Reduction Strategies in Retrieval-Augmented Generation (RAG)

1. Introduction

Large Language Models (LLMs) have demonstrated remarkable capabilities in natural language understanding and generation. However, they are prone to hallucinations, generating fluent but factually incorrect or unsupported content. Retrieval-Augmented Generation (RAG) has emerged as a prominent paradigm to mitigate hallucinations by grounding model outputs in external knowledge sources. Despite its effectiveness, hallucinations persist due to retrieval errors, reasoning gaps, and generation misalignment. This survey reviews key strategies proposed in the literature to reduce hallucinations in RAG systems.

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2. Hallucination in RAG: Problem Characterization

Hallucinations in RAG systems can be broadly categorized into:

Retrieval hallucinations: Incorrect or irrelevant documents retrieved.

Contextual hallucinations: Model ignores or misinterprets retrieved evidence.

Fabrication hallucinations: Model generates content not supported by retrieved sources.

Studies such as Ji et al. (2023) and Huang et al. (2023) emphasize that hallucinations in RAG are not solely a generation problem but arise from interactions across retrieval, fusion, and decoding stages.

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3. Retrieval-Level Hallucination Reduction Strategies

3.1 Improved Document Retrieval

Early RAG implementations (Lewis et al., 2020) rely on dense passage retrieval (DPR). Subsequent works show that retrieval quality strongly correlates with hallucination rates. Strategies include:

Hybrid retrieval combining sparse (BM25) and dense retrievers (Izacard et al., 2022).

Query rewriting and decomposition to improve recall and relevance (Ma et al., 2023).

Multi-hop retrieval for complex questions requiring chained evidence.

3.2 Retrieval Verification and Filtering

To avoid injecting noisy context:

Re-ranking using cross-encoders (Nogueira & Cho, 2019)

Confidence-based filtering of retrieved documents

Redundancy-aware selection to avoid contradictory sources

These approaches reduce hallucinations caused by irrelevant or low-quality evidence.

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4. Context Fusion and Prompting Strategies

4.1 Evidence-Aware Prompting

Prompt engineering plays a crucial role in hallucination reduction. Evidence-aware prompts instruct the model to:

Use only retrieved context

Explicitly cite or quote evidence

Abstain when information is insufficient

Studies show that constrained prompts significantly reduce unsupported claims (Mialon et al., 2023).

4.2 Structured Context Representation

Rather than raw text concatenation, structured formats such as:

Bullet-point summaries

Key-value extractions

Graph-based representations

help models better align generation with evidence (Liu et al., 2023).

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5. Generation-Level Control Mechanisms

5.1 Faithfulness-Aware Decoding

Decoding strategies influence hallucinations:

Lower temperature and nucleus sampling reduce creative fabrications.

Evidence-conditioned decoding penalizes tokens not grounded in retrieved text.

Approaches such as contrastive decoding and fact-aware decoding have shown measurable hallucination reduction.

5.2 Attribution and Citation Enforcement

Requiring models to generate citations for each claim (e.g., RET-ATTR, Cite-then-Generate) improves factual grounding and allows post-generation verification.

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6. Post-Generation Verification and Self-Correction

6.1 Answer Verification

Several works propose verifying generated answers against retrieved documents using:

Natural language inference (NLI) models

Answer consistency checks

Secondary LLM-based critics

If contradictions are detected, regeneration is triggered.

6.2 Self-Reflection and Critique

Recent studies show that self-reflection, where the model critiques its own output, can reduce hallucinations. Techniques such as Self-RAG and Reflexion iteratively refine answers by checking evidence alignment.

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7. Training-Time Approaches

7.1 Fine-Tuning with Faithfulness Objectives

Fine-tuning LLMs on datasets labeled for factual correctness improves grounding. Loss functions penalizing unsupported generations have been proposed.

7.2 Reinforcement Learning from Human Feedback (RLHF)

RLHF has been used to discourage hallucinations by rewarding faithful, evidence-backed responses. However, it is resource-intensive and domain-specific.

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8. Evaluation of Hallucinations in RAG

Evaluating hallucination remains challenging. Common approaches include:

Human evaluation for factual consistency

Automatic metrics using entailment and overlap scores

Benchmark datasets such as TruthfulQA and RAG-specific faithfulness benchmarks

Recent work emphasizes task-specific evaluation rather than generic factuality scores.

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9. Open Challenges and Future Directions

Despite progress, several challenges remain:

Robust retrieval in low-resource or rapidly changing domains

Handling contradictory evidence

Balancing informativeness and abstention

Scalable evaluation of hallucinations

Future research is moving toward end-to-end retriever-generator co-training, uncertainty-aware generation, and neuro-symbolic grounding.

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10. Conclusion

Hallucination reduction in RAG systems requires a holistic approach spanning retrieval, prompting, generation, and verification. While no single strategy fully eliminates hallucinations, combining retrieval quality improvements, evidence-aware generation, and post-hoc verification yields substantial gains. As RAG systems become foundational in enterprise and safety-critical applications, hallucination mitigation remains a central research priority.

Key References (Indicative)

Lewis et al., 2020 – Retrieval-Augmented Generation for Knowledge-Intensive NLP

Ji et al., 2023 – Survey of Hallucination in NLP

Izacard et al., 2022 – Few-shot Learning with Retrieval-Augmented Models

Mialon et al., 2023 – Augmented Language Models: A Survey

Huang et al., 2023 – A Survey on Hallucination in Large Language Models